# SGD, noise, symmetries, and implicit bias

Student notes

Here *IB* means *implicit bias*: the preference for some predictors among many that fit the training data. We first study a factorized predictor, then charge contraction, and finally the geometric weighting of predictors and its inverse design.

## 1 Underdetermined regression and a factorized predictor

Let $x_i \in \mathbb{R}^d$, $y_i = w_\star^\mathsf{T} x_i$, and consider

$$\mathcal{L}(w) = \frac{1}{N}\sum_{i=1}^{N}(y_i - w^\mathsf{T} x_i)^2. \tag{1}$$

For the example in the notes, $d = 42$ and $N = 20$. With $X = [x_1, \dots, x_N] \in \mathbb{R}^{d\times N}$, interpolation means $X^\mathsf{T} w = y$. Since $\operatorname{rank} X \le 20 < 42$, the solutions form an affine space

$$\{w : X^\mathsf{T} w = y\} = w_\star + \ker X^\mathsf{T}.$$

Thus fitting the data does not uniquely determine the predictor. Gradient descent updates $w_{k+1} = w_k - \eta \nabla_w \mathcal{L}(w_k)$, where $\eta > 0$ is the learning rate.

Now parameterize $w = u \odot v$, where $\odot$ denotes coordinatewise multiplication. The prediction becomes

$$\widehat{y} = (u \odot v)^\mathsf{T} x.$$

To understand the geometry, focus first on one scalar pair $\theta = (u, v)$ and write

$$w = h(u, v) = uv, \qquad L(\theta) = \ell(h(\theta)). \tag{2}$$

Here $\ell$ is a scalar loss. In the vector model, the same calculation applies to each pair $(u_j, v_j)$, with $g_j = \partial\mathcal{L}/\partial w_j$ in place of $\ell'(h)$.

## 2 A symmetry and its conserved charge

The transformations

$$\Phi_\alpha(u, v) = (e^\alpha u, e^{-\alpha} v) \tag{3}$$

preserve the predictor because $h(\Phi_\alpha(u, v)) = uv$. Their infinitesimal generator is

$$V(u, v) = \frac{\mathrm{d}}{\mathrm{d}\alpha}\Phi_\alpha(u, v)\Big|_{\alpha=0} = (u, -v).$$

This is a gradient field:

$$V = \nabla C, \qquad C(u, v) = \frac{1}{2}(u^2 - v^2), \qquad \nabla h = (v, u), \qquad \nabla C \cdot \nabla h = 0. \tag{4}$$

We call $C$ the *charge*. Under gradient flow,

$$\dot{\theta} = -\nabla_\theta L = -\ell'(h)\nabla h, \qquad \dot{u} = -\ell'(h)v, \quad \dot{v} = -\ell'(h)u. \tag{5}$$

Therefore

$$\frac{\mathrm{d}C}{\mathrm{d}t} = u\dot{u} - v\dot{v} = -u\ell'(h)v + v\ell'(h)u = 0. \tag{6}$$

The symmetry preserves $h$, while gradient flow preserves $C$. More generally, any smooth pair $(h, C)$ satisfying $\nabla C \cdot \nabla h = 0$ has this conservation property for a loss $L = \ell(h)$.

# 3 Discretization changes the charge

## 3.1 An exact calculation for $h = uv$

Let $g_k$ be the derivative of the loss with respect to the scalar predictor. It can be a full-batch derivative or a random mini-batch derivative. Both factors must use the same $g_k$ and the same pre-update state:

$$u_{k+1} = u_k - \eta g_k v_k, \qquad v_{k+1} = v_k - \eta g_k u_k. \tag{7}$$

Expanding the squares gives

$$C_{k+1} = \frac{1}{2}\left[(u_k - \eta g_k v_k)^2 - (v_k - \eta g_k u_k)^2\right] \tag{8}$$

$$= \frac{1}{2}(u_k^2 - v_k^2)(1 - \eta^2 g_k^2) = \boxed{(1 - \eta^2 g_k^2)C_k.} \tag{9}$$

The terms linear in $\eta$ cancel. The charge changes at second order, even for deterministic gradient descent.

One-step contraction of its magnitude requires $|1 - \eta^2 g_k^2| < 1$, equivalently $0 < \eta^2 g_k^2 < 2$. For $0 < \eta^2 g_k^2 < 1$, contraction also preserves its sign. Over many steps,

$$C_k = C_0 \prod_{j=0}^{k-1}(1 - \eta^2 g_j^2). \tag{10}$$

Having every factor smaller than one is not enough by itself to prove $C_k \to 0$: the product must vanish. For example, if $0 \le \eta^2 g_j^2 \le a < 1$ and $\sum_j \eta^2 g_j^2 = \infty$, then $|C_k| \le |C_0| \exp(-\sum_{j<k} \eta^2 g_j^2) \to 0$. If the gradients vanish too rapidly, a nonzero charge can remain.

For mini-batch SGD, conditional on $\theta_k$ and assuming $\mathbb{E}[g_k \mid \theta_k] = \ell'(h_k)$,

$$\mathbb{E}[C_{k+1} - C_k \mid \theta_k] = -\eta^2 C_k \left(\ell'(h_k)^2 + \mathrm{Var}(g_k \mid \theta_k)\right). \tag{11}$$

This displays how gradient fluctuations contribute to the second-order charge drift. It is a conditional drift identity, not by itself a mean-square convergence proof.

## 3.2 The two orders in a general parameterization

Take smooth $h, C$ with $\nabla C \cdot \nabla h = 0$ and $\theta_{k+1} = \theta_k - \eta g_k \nabla h(\theta_k)$. Taylor expansion gives, with derivatives evaluated at $\theta_k$,

$$h(\theta_{k+1}) - h(\theta_k) = -\eta g_k \|\nabla h\|^2 + O(\eta^2), \tag{12}$$

$$C(\theta_{k+1}) - C(\theta_k) = \frac{\eta^2 g_k^2}{2} \underbrace{\nabla h^\mathsf{T} \nabla^2 C \, \nabla h}_{Q(\theta_k)} + O(\eta^3). \tag{13}$$

The predictor changes at first order; the charge changes at second order. These expansions assume bounded derivatives and increments in the region considered. For $h = uv$ and $C = (u^2 - v^2)/2$,

$$\nabla^2 C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q = v^2 - u^2 = -2C,$$

which recovers the exact update (9). For a general charge, contraction means motion toward zero: the sign of $CQ$ matters, rather than the sign of $Q$ alone.

## 4 Controlling the second-order charge drift

Suppose the natural curvature term $Q$ does not produce the desired contraction. The notes propose a correction along the charge gradient that leaves the leading predictor update unchanged.

At a regular point, a vector $\xi$ cannot be orthogonal to both $\nabla h$ and $\nabla C$ in two dimensions unless $\xi = 0$. Instead, put the correction at second order. To make its sign and scaling explicit, define

$$\theta_{k+1} = \theta_k - \eta g_k \nabla h + \eta^2 g_k^2 \xi(\theta_k). \tag{14}$$

Then

$$\Delta C = \eta^2 g_k^2 \left( \nabla C \cdot \xi + \frac{1}{2} Q \right) + O(\eta^3), \tag{15}$$

$$\Delta h = -\eta g_k \left\| \nabla h \right\|^2 + \eta^2 g_k^2 \left( \nabla h \cdot \xi + \frac{1}{2} \nabla h^{\mathsf{T}} \nabla^2 h \, \nabla h \right) + O(\eta^3). \tag{16}$$

Choose $\xi = q(\theta) \nabla C$. Orthogonality gives $\nabla h \cdot \xi = 0$, so the correction adds no second-order *linear* contribution to $\Delta h$. To impose

$$\Delta C = -\frac{\eta^2 g_k^2}{2} \kappa C + O(\eta^3), \qquad \kappa > 0,$$

solve $q \left\| \nabla C \right\|^2 + Q/2 = -\kappa C/2$:

$$\boxed{\xi(\theta) = -\frac{Q(\theta) + \kappa C(\theta)}{2 \left\| \nabla C(\theta) \right\|^2} \nabla C(\theta).} \tag{17}$$

This formula applies where $\nabla C \neq 0$. It controls the leading second-order term; finite-step convergence also depends on the remainder and the learning rate. The specified $\xi$ is a deterministic correction, not automatically a zero-mean noise.

**Connection with the Itô calculation (for completeness).** The continuous stochastic counterpart of a quadratic correction is quadratic variation. For the Itô equation

$$\mathrm{d}\theta = b(\theta)\,\mathrm{d}t + G(\theta)\,\mathrm{d}W_t, \qquad A = GG^{\mathsf{T}},$$

the change of any smooth function $f$ is

$$\mathrm{d}f = \left( \nabla f \cdot b + \frac{1}{2} \operatorname{tr}(A \nabla^2 f) \right) \mathrm{d}t + \nabla f^{\mathsf{T}} G \, \mathrm{d}W_t. \tag{18}$$

For $b = -\ell'(h) \nabla h$, the gradient-drift contribution to $\mathrm{d}C$ vanishes. The Hessian term remains. Unlike a smooth deterministic increment, a stochastic increment is of order $\sqrt{\mathrm{d}t}$, so its quadratic contribution is of order $\mathrm{d}t$.

# 5 From parameter volume to predictor weights

## 5.1 The coordinate change

Write the values of the predictor and charge as

$$\Psi(\theta) = (h(\theta), C(\theta)) = (w, c). \tag{19}$$

On a regular invertible patch, define

$$J(\theta) = |\det D\Psi(\theta)| = |h_u C_v - h_v C_u|, \qquad \Gamma(w,c) = \frac{1}{J(\theta(w,c))}. \tag{20}$$

For orthogonal gradients, $J = \|\nabla h\|\,\|\nabla C\|$. The change-of-variables formula is

$$\boxed{\mathrm{d}u\,\mathrm{d}v = \Gamma(w,c)\,\mathrm{d}w\,\mathrm{d}c.} \tag{21}$$

Thus an integral over parameter volume becomes

$$\int_{\mathcal{D}} f(h(\theta))\,\mathrm{d}\theta = \iint_{\Psi(\mathcal{D})} f(w)\Gamma(w,c)\,\mathrm{d}w\,\mathrm{d}c.$$

The factor $\Gamma$ measures the parameter volume associated with a unit area in predictor–charge coordinates.

If there are several inverse branches, sum $1/J$ over the accessible branches when transforming unweighted volume. For a parameter density $p(\theta)$, the transformed density is instead

$$P(w,c) = \sum_{\theta_b:\,\Psi(\theta_b)=(w,c)} \frac{p(\theta_b)}{J(\theta_b)}. \tag{22}$$

Branch populations therefore matter.

## 5.2 IB derivation

Restricting the charge to zero induces a geometric weight on the predictor:

$$\int f(h(\theta))\,\delta(C(\theta))\,\mathrm{d}\theta = \int_{C=0} \frac{f(h(\theta))}{\|\nabla C(\theta)\|}\,\mathrm{d}s = \int f(w)\,\Pi_0(w)\,\mathrm{d}w.$$

Here $f$ is a test function. In the first integral, the Dirac delta $\delta(C(\theta))$ restricts integration to the curve $C = 0$. In the second integral, $\mathrm{d}s$ denotes distance along this curve. The factor $1/\|\nabla C(\theta)\|$ accounts for the conversion from charge intervals to distances perpendicular to the curve: where the charge changes more slowly, a fixed small charge interval corresponds to more parameter volume.

The last equality uses the predictor value $w = h(\theta)$ as the integration variable. The resulting weight $\Pi_0(w)$ collects the contributions of all accessible points with predictor $w$ and charge zero:

$$\Pi_0(w) = \sum_{\substack{\theta_b:\,h(\theta_b)=w \\ C(\theta_b)=0}} \frac{1}{J(\theta_b)}, \qquad J(\theta) = \left|\det \frac{\partial(h,C)}{\partial(u,v)}\right|.$$

Thus larger $\Pi_0(w)$ means greater geometric weight for that predictor value under the constrained parameter measure. The formulas apply at regular points where $\nabla C \neq 0$ and $J \neq 0$; singular points require a limiting treatment.

**The implicit-bias potential** Rewrite the predictor weight as

$$\Pi_0(w)e^{-\beta E(w)} = e^{-\beta[E(w)+\beta^{-1}\mathcal{R}_{\mathrm{IB}}(w)]}, \qquad \boxed{\mathcal{R}_{\mathrm{IB}}(w) = -\log \Pi_0(w).} \tag{23}$$

An additive constant in $\mathcal{R}_{\mathrm{IB}}$ only changes normalization. Large geometric weight $\Pi_0$ corresponds to a low effective potential. The reference units inside logarithms are fixed.

### 5.3 Example: $h = uv$

For the pair (4),

$$D\Psi = \begin{pmatrix} v & u \\ u & -v \end{pmatrix}, \qquad J = u^2 + v^2 = 2\sqrt{w^2 + c^2}. \tag{24}$$

On a single inverse branch,

$$\Gamma(w, c) = \frac{1}{2\sqrt{w^2 + c^2}}, \qquad \Gamma(w, 0) = \frac{1}{2|w|} \quad (w \neq 0). \tag{25}$$

With constant branch multiplicity and uniform reference weight,

$$\Pi_0(w) \propto \frac{1}{|w|}, \qquad \mathcal{R}_{\mathrm{IB}}(w) = \log |w| + \text{constant}. \tag{26}$$

Hence $\log |w| \to -\infty$ as $w \to 0$: the formal geometric weighting favors small predictor values. Applied coordinatewise, it gives $\mathcal{R}_{\mathrm{IB}}(w) = \sum_j \log |w_j| + \text{constant}$, a sparsity-favoring expression.

At $w = c = 0$, however, the Jacobian vanishes. The weight $1/|w|$ is not locally integrable there when the other factors remain nonzero and finite. Thus (26) describes a singular limiting preference, not an automatically normalized probability law or a proof of sparse convergence.

## 6 Inverse design with holomorphic functions

### 6.1 Orthogonality from the Cauchy–Riemann equations

Introduce $z = u + iv$ and a holomorphic function

$$F(z) = C(u, v) + i\, h(u, v). \tag{27}$$

Its real and imaginary parts obey

$$C_u = h_v, \qquad C_v = -h_u, \qquad \Delta C = \Delta h = 0. \tag{28}$$

Thus they are harmonic conjugates, and

$$\nabla h \cdot \nabla C = h_u C_u + h_v C_v = (-C_v)C_u + C_u C_v = 0.$$

The orthogonality needed for charge conservation is built into this parameterization.

The complex derivative is

$$F'(z) = C_u + i h_u = C_u - i C_v. \tag{29}$$

Consequently,

$$\boxed{|F'(z)|^2 = C_u^2 + C_v^2 = \|\nabla C\|^2 = \|\nabla h\|^2 = \left|\det \frac{\partial(h, C)}{\partial(u, v)}\right|.} \tag{30}$$

There is no factor of two: the complex derivative uses one coordinate derivative, not the sum of the $u$ and $v$ derivatives. The volume factor is therefore $1/|F'(z)|^2$ on each regular branch.

### 6.2 Placing enhanced geometric weights

For the selected charge $C = 0$, the geometric predictor weight is formally

$$\Pi_0(w) = \sum_{\substack{z:\ \Re F(z)=0 \\ \Im F(z)=w}} \frac{1}{|F'(z)|^2}, \tag{31}$$

with accessible branches understood. A zero $\lambda$ of $F'$ is a critical point of the coordinate map. Its predictor value is $w_\lambda = \Im F(\lambda)$; it lies on the selected level only if $\Re F(\lambda) = 0$.

The design idea is to choose

$$F'(z) = \alpha \prod_{j=1}^{m} (z - \lambda_j), \qquad \alpha \neq 0, \qquad F(z) = \int F'(z)\,\mathrm{d}z + K. \tag{32}$$

The zeros $\lambda_j$ are points in *parameter space*; their images $F(\lambda_j)$ determine the favored predictor values and charge levels. The formula for the weight is used away from the critical points and examined by a limit near them. Holomorphicity guarantees orthogonality, but charge contraction remains a separate dynamical condition.

### 6.3 Example 1: the product predictor

Choose $F'(z) = z$ and $F(z) = z^2/2$. Expanding gives

$$F(u + iv) = \frac{1}{2}(u^2 - v^2) + iuv.$$

Thus $C = (u^2 - v^2)/2$, $h = uv$, and $|F'|^2 = u^2 + v^2$, recovering the earlier example. The critical point $z = 0$ maps to $(C, h) = (0, 0)$.

### 6.4 Example 2: two symmetric predictor values

Choose critical points $\lambda_\pm = \pm i$, so that

$$F'(z) = (z - i)(z + i) = z^2 + 1, \qquad F(z) = \frac{z^3}{3} + z. \tag{33}$$

The real and imaginary parts are

$$\boxed{C(u, v) = \frac{u}{3}(u^2 - 3v^2 + 3), \qquad h(u, v) = \frac{v}{3}(3u^2 - v^2 + 3).} \tag{34}$$

Indeed,

$$F(i) = \frac{2i}{3}, \qquad F(-i) = -\frac{2i}{3}.$$

Both critical points lie on $C = 0$, and their predictor values are $w = \pm 2/3$. Their coarea weights become singular as these critical values are approached along the relevant branches.

If the desired predictor values are instead $\pm 1$, multiply $F$ by $3/2$:

$$\widetilde{F}(z) = \frac{z^3}{2} + \frac{3}{2}z, \qquad \widetilde{C} = \frac{u}{2}(u^2 - 3v^2 + 3), \qquad \widetilde{h} = \frac{v}{2}(3u^2 - v^2 + 3). \tag{35}$$

Then $\widetilde{F}(\pm i) = \pm i$. The same scaling must be applied to both real and imaginary parts to preserve the holomorphic construction. This places enhanced geometric weights at the two predictor values; whether training reaches them depends on the charge dynamics and the distribution generated by the updates.